Differential equations
Part A & B: Intro and Slope Field

3unit
Selective

Q 1 ASCHAM

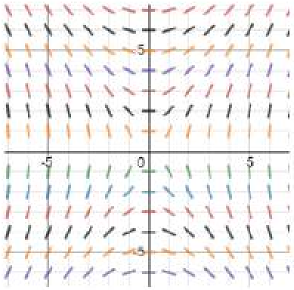
The slope field below could represent which of the following differential equations?

A. $\frac{dy}{dx} = \frac{2x}{y}$

B. $\frac{dy}{dx} = \frac{2y}{x}$

C. $\frac{dy}{dx} = \frac{x^2}{y^2}$

D. $\frac{dy}{dx} = \frac{y^2}{x^2}$



Q 2 BAULKHAM

The slope field for a differential equation is shown below.

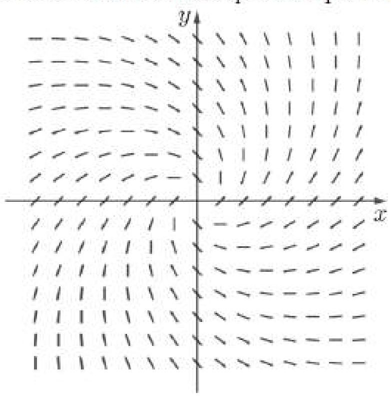
Which of the following could be the differential equation represented?

A. $\frac{dy}{dx} = \frac{x+y}{x-y}$

B. $\frac{dy}{dx} = \frac{y+x}{y-x}$

C. $\frac{dy}{dx} = \frac{x-y}{x+y}$

D. $\frac{dy}{dx} = \frac{y-x}{y+x}$



From the slope field, $\frac{dy}{dx} = 0$ when $y = -x$ and $\frac{dy}{dx}$ is undefined when $y = x$.

Q 3 BLACKTOWN

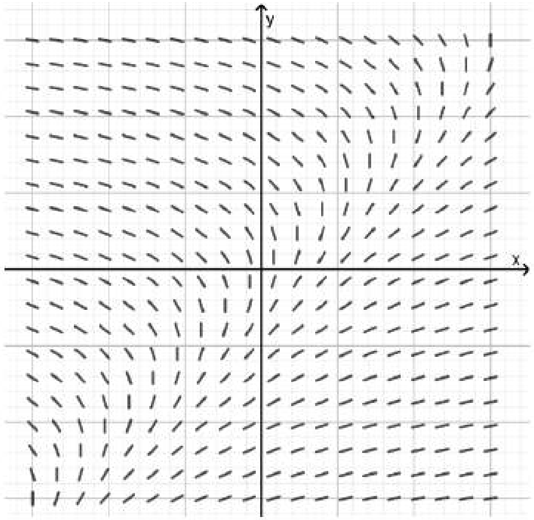
The differential equation that is best represented by the direction field below is

A. $\frac{dy}{dx} = x - y$

B. $\frac{dy}{dx} = \frac{1}{x-y}$

C. $\frac{dy}{dx} = y - x$

D. $\frac{dy}{dx} = \frac{1}{y-x}$



The vertical line segments on the line $y = x$, so option B and D are possibilities. Positive gradients for $x = 0$ and $y < 0$ then gives option B.

Q 4 BONNYRIGG

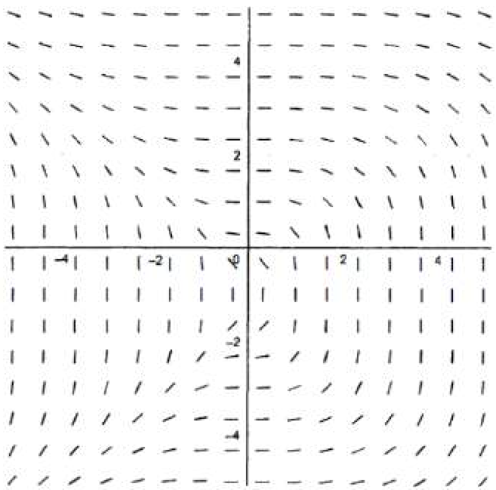
Shown above is the slope field for which differential equation?

A. $\frac{dy}{dx} = \frac{3x^2}{(y+1)^3}$

B. $\frac{dy}{dx} = \frac{3x}{(y+1)^2}$

C. $\frac{dy}{dx} = -\frac{3x^2}{(y+1)^3}$

D. $\frac{dy}{dx} = -\frac{3x}{(y+1)^2}$



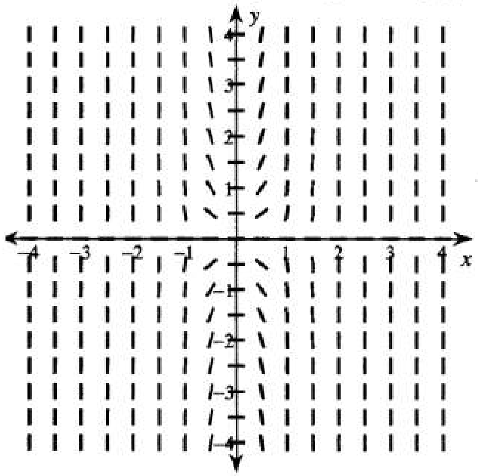
Q 5 BONNYRIGG

The slope field for the differential equation $\frac{dy}{dx} = 12x^3y$ is shown below.

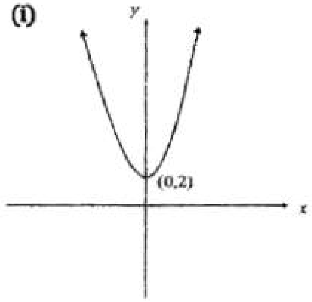
(i) Sketch the solution curve passing through $(0, 2)$. **1**

(You do not need to draw the slope field).

(ii) Find the solution to the differential equation given that $y(0) = 2$. **2**



(i)



(ii) $\frac{dy}{dx} = 12x^3y$

$\int \frac{1}{y}\,dy = \int 12x^3\,dx$

$\ln|y| = 3x^4 + C$

$x = 0, y = 2, C = \ln 2$

$\ln|y| = 3x^4 + \ln 2$

$|y| = e^{3x^4 + \ln 2}$

$\therefore y = 2e^{3x^4}$ since $y > 0\ \forall x \in \mathbb{R}$ (see diagram in (c)(i))

Q 6 CHELTENHAM GIRLS

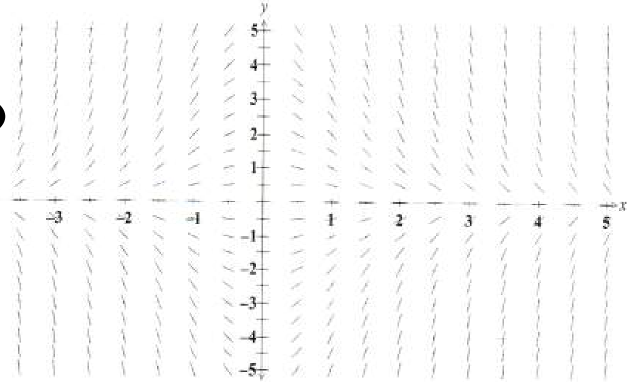
Which of the following differential equations could be represented by the slope field diagram below?

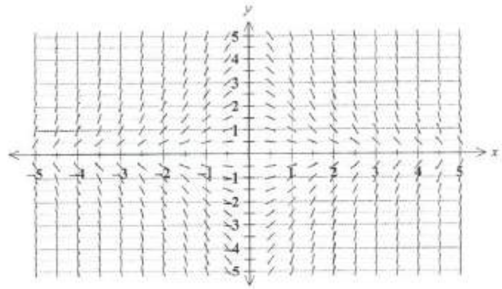
A. $y' = -xy$

B. $y' = xy$

C. $y' = -x^2y$

D. $y' = x^2y$





The slope in the 1st and 3rd quadrants is always negative.
The slope in the 2nd and 4th quadrants is always positive.
This rules out options B, C and D.

Q 7 GRAFTON

The slope field for a first order differential equation is shown.

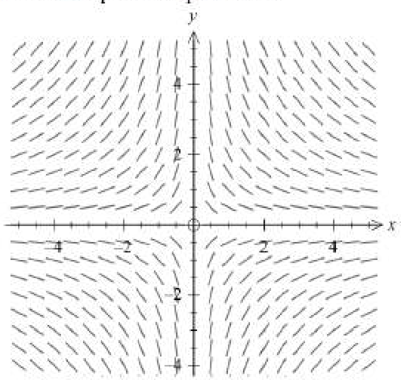
What could be the differential equation represented?

(A) $\frac{dy}{dx} = -\frac{x}{y}$

(B) $\frac{dy}{dx} = \frac{x}{y}$

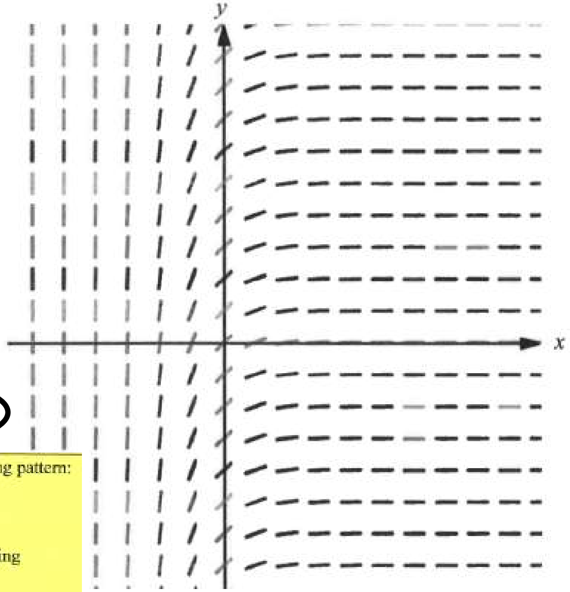
(C) $\frac{dy}{dx} = -\frac{y}{x}$

(D) $\frac{dy}{dx} = \frac{y}{x}$



Q 8 CSSA

Which of the following could be the differential equation represented by the slope field below?



(A) $\dfrac{dy}{dx} = \ln x$

(B) $\dfrac{dy}{dx} = -\ln x$

(C) $\dfrac{dy}{dx} = e^{x}$

(D) $\dfrac{dy}{dx} = e^{-x}$

The slope field follows the following pattern:

as $x \to -\infty, \dfrac{dy}{dx} \to \infty$

for $x \in (-\infty, \infty), \dfrac{dy}{dx} > 0$ and decreasing

as $x \to \infty, \dfrac{dy}{dx} \to 0$

$\therefore \dfrac{dy}{dx} = e^{-x}$

Hence (D)

Q 9 INDEPENDENT

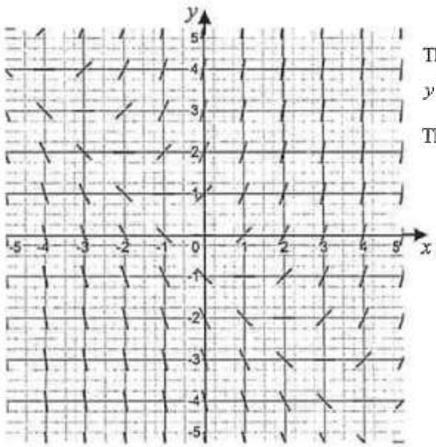
Which differential equation is represented by the following slope field?

(A) $\dfrac{dy}{dx} = x + y$

(B) $\dfrac{dy}{dx} = x - y$

(C) $\dfrac{dy}{dx} = \dfrac{x}{y}$

(D) $\dfrac{dy}{dx} = \dfrac{y}{x}$



The slope segments along the isocline $y = -x$ suggest $\dfrac{dy}{dx} = 0$ when $y = -x$.

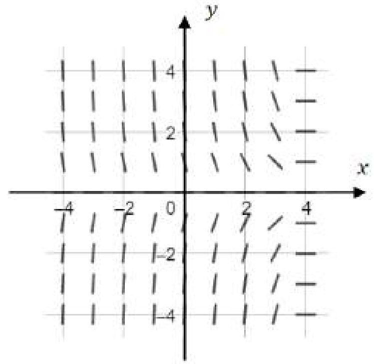
This is true only for A.

Q10 HUNTERS HILL

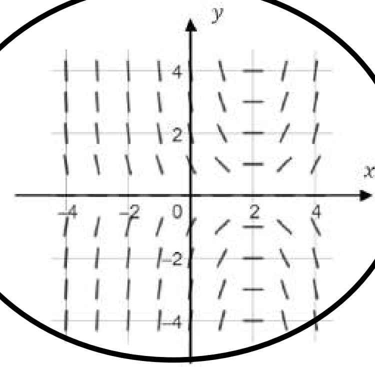
A differential equation is given by $y' = xy - 2y$

Which of the following slope fields best represents the differential equation?

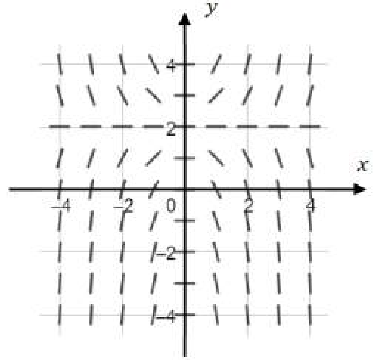
(A)



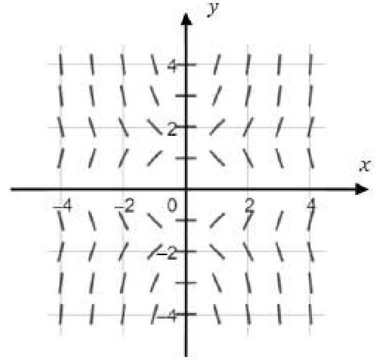
(B)



(C)



(D)



$y' = y(x-2)$ $\quad$ $x=2 \Rightarrow y'=2$

Q11 JAMES RUSE

Which one of the following is a first-order linear differential equation?

(A) $xy' = 14x^2 + 9y$

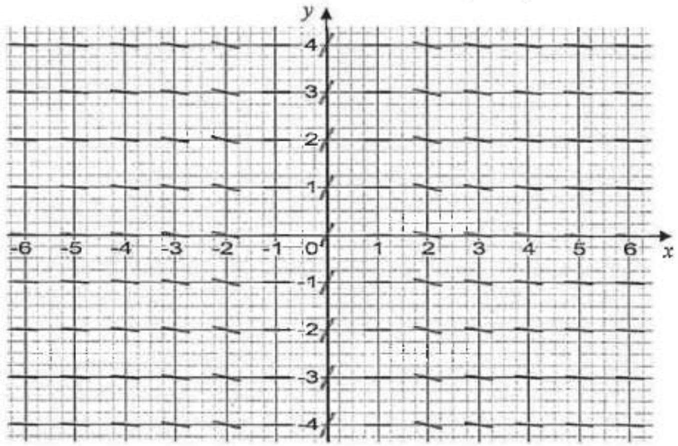
(B) $y'' + 2y' - 8y = 0$

(C) $y^2 + y + x = 0$

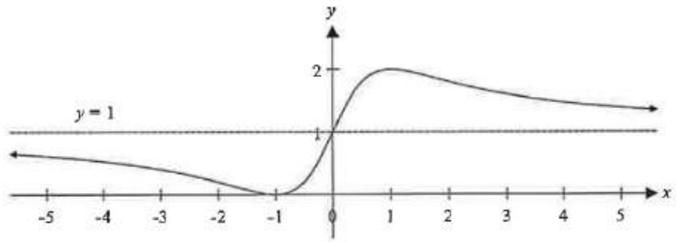
(D) $y'y - 2x^2y^2 = 14x$

Q12 INDEPENDENT

The slope field for the differential equation $\frac{dy}{dx}=\frac{2(1-x^2)}{(1+x^2)^2}$ is given below. **3**



Draw a sketch of the solution curve $y=f(x)$ that passes through $(-1,0)$ and describe its behaviour as $x\to\pm\infty$.

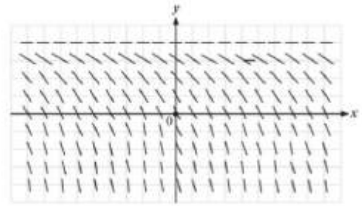


$y=1$ is a horizontal asymptote with $y\to1^{+}$ as $x\to+\infty$, and $y\to1^{-}$ as $x\to-\infty$.

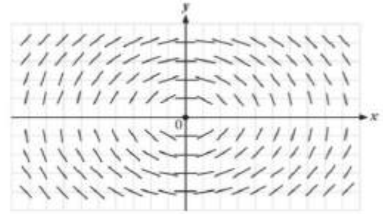
Q13 JAMES RUSE

The slope field that represents the differential equation $\frac{dy}{dx}=\frac{x}{2y}$ is which of the following?

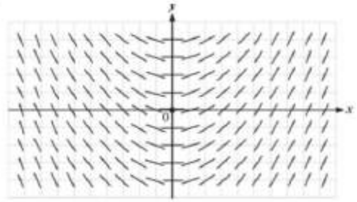
(A)



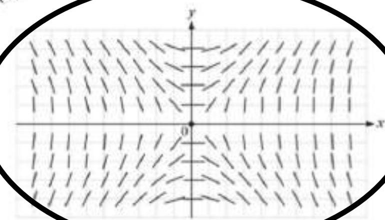
(B)



(C)

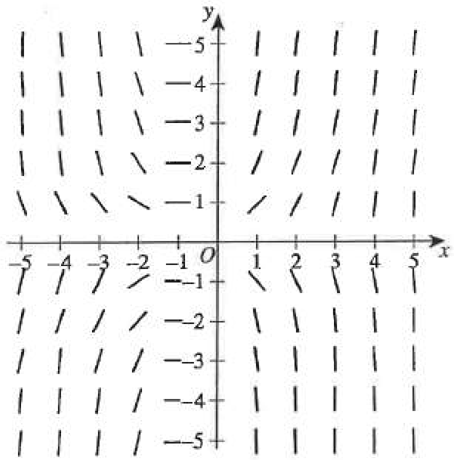


(D)



Q14 NEAP

The direction (slope) field for a first order differential equation is shown.



At $(0, 0)$, $\frac{dy}{dx}=0$ and so **A** is incorrect.

At $(-1, 1)$, $\frac{dy}{dx}=0$ and so **B** and **D** are incorrect.

Which of the following could be the differential equation represented?

(A) $\frac{dy}{dx}=(x+1)^3$

(B) $\frac{dy}{dx}=x(y+1)$

(C) $\frac{dy}{dx}=(x+1)y$

(D) $\frac{dy}{dx}=(x-1)y$

Q15

NEWINGTON

The slope field for a first order differential equation is shown below.

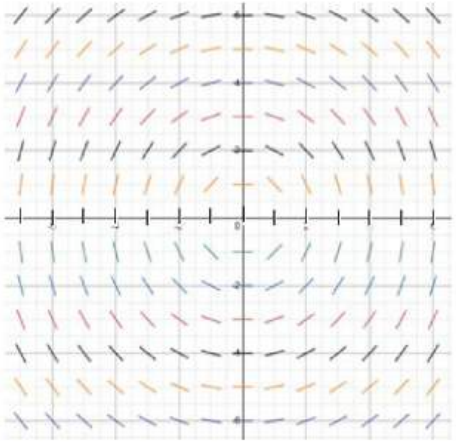
Which of the following could be the differential equation represented?

(A) $\frac{dy}{dx} = \frac{x}{y}$

(B) $\frac{dy}{dx} = \frac{-x}{y}$

(C) $\frac{dy}{dx} = xy$

(D) $\frac{dy}{dx} = -xy$

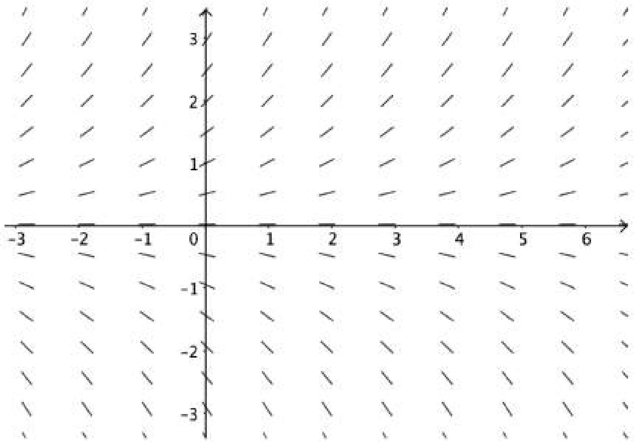


By considering slopes at different points of cartesian plane and testing with each differential equation

Q16

NORMANHURST

Consider the slope field shown below.



What is a possible differential equation for the slope field?

(A) $\frac{dy}{dx} = -\frac{x}{2}$ (B) $\frac{dy}{dx} = \frac{x}{2}$ (C) $\frac{dy}{dx} = -\frac{y}{2}$ (D) $\frac{dy}{dx} = \frac{y}{2}$

Q17

PENRITH

A large tank initially holds 2500L of water in which 100kg of salt is dissolved. A solution containing 4kg of salt per litre flows into the tank at a rate of 10L per minute. The mixture is stirred continuously and flows out of the tank through a hole at a rate of 14L per minute.

A differential equation for Q, the number of kilograms of salt in the tank after t minutes, is given by

A. $\frac{dQ}{dt} = 40 - \frac{7Q}{2(625-t)}$

B. $\frac{dQ}{dt} = 40 + \frac{7Q}{2(625-t)}$

C. $\frac{dQ}{dt} = 40 - \frac{7Q}{2(625+t)}$

D. $\frac{dQ}{dt} = 40 + \frac{7Q}{2(625+t)}$

$\frac{dQ}{dt} = 4\text{kg} \times 10\text{ L/min} - \frac{Q \cancel{2500} \times 14\text{L/min}}{(2500 + (10\text{L/min} - 14\text{L/min})t)}$

$= 40 - \frac{14Q}{(2500-4t)}$

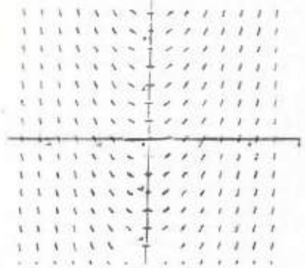
$= 40 - \frac{7Q}{2(625-t)}$ A

Q18

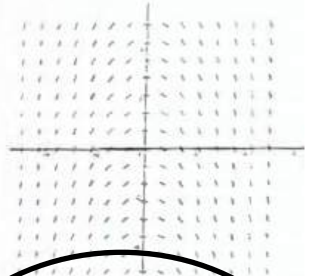
PENRITH

Which of the following slope field best represents the differential equation $y' = x + y$

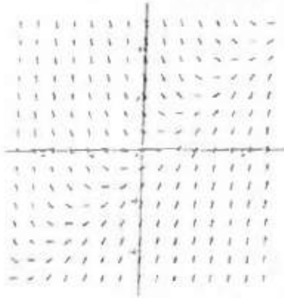
A.



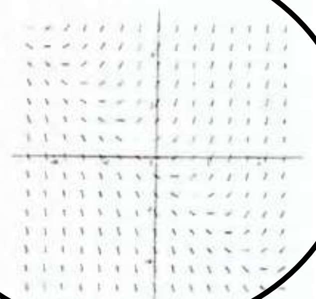
B.



C.



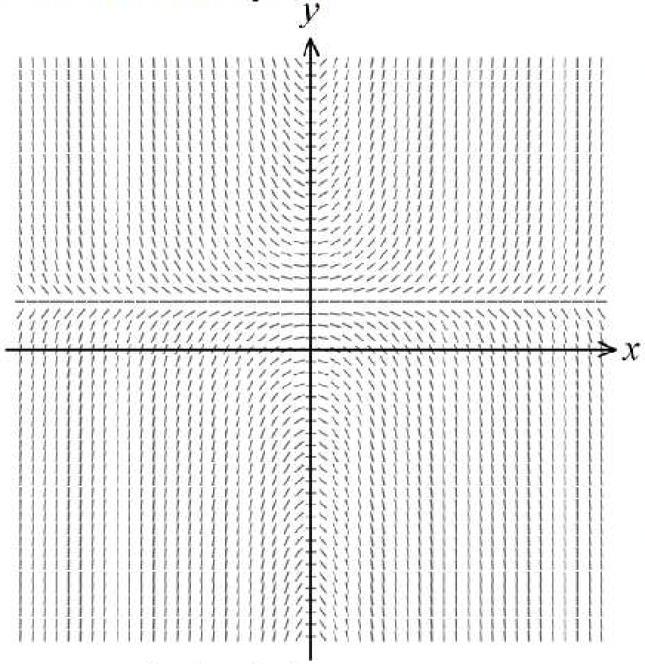
D.



Q19 SYDNEY BOYS

The direction (slope) field for a certain first order differential equation is shown below.

Which of the following below is the differential equation?



A. $\frac{dy}{dx} = x^2 + y^2$

B. $\frac{dy}{dx} = x^2 - y^2$

C. $\frac{dy}{dx} = xy - x$

D. $\frac{dy}{dx} = y^2 - x^2$

The diagram indicates that $\frac{dy}{dx} = 0$ if $x = 0$

$\therefore$ A, B, D do not guarantee this

Also for C $\frac{dy}{dx} = 0$ if $y = 1$

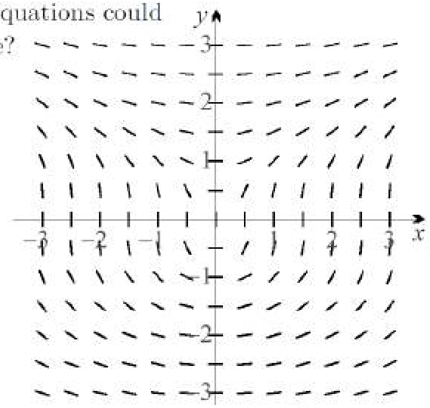
$\frac{dy}{dx} = xy - x$
$= x(y-1)$

C is the only diagram meeting these requirements.

C

Q20 SYDNEY GRAMMAR

Which of the following differential equations could produce the slope field shown above?



(A) $\frac{dy}{dx} = \frac{x}{y^2}$

(B) $\frac{dy}{dx} = xy$

(C) $\frac{dy}{dx} = \frac{x}{y}$

(D) $\frac{dy}{dx} = x^2 y$

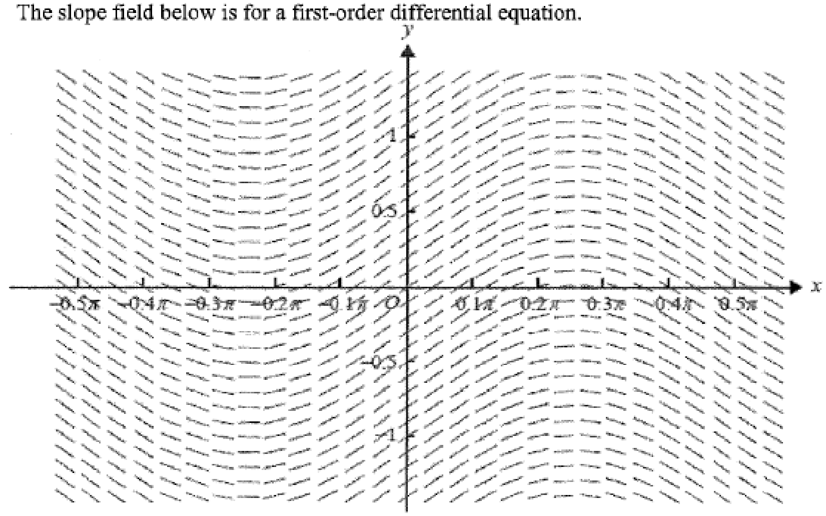
$x=0$ $\frac{dy}{dx}=0$ $y=0$ Gradients vertical hence not (B) or (D)

1^{st} quad $\frac{dy}{dx}>0$ 2^{nd} quad $\frac{dy}{dx}<0$ 3^{rd} quad $\frac{dy}{dx}<0$ hence not (C)

hence (A)

Q21 SYDNEY TECH

The slope field below is for a first-order differential equation.



Which of the following is a possible differential equation?

(A) $\frac{dy}{dx} = \sin 2x$

(B) $\frac{dy}{dx} = \sin 2y$

(C) $\frac{dy}{dx} = \cos 2x$

(D) $\frac{dy}{dx} = \cos 2y$

Q 1 Ascham

Find the particular solutions to the differential equation $\frac{dy}{dx}=\frac{2x-1}{y^2}$ given $y=\sqrt[3]{3}$ when $x=0$. **3**

1) $\frac{dy}{dx}=\frac{2x-1}{y^2}$

$\int y^2\,dy=\int(2x-1)\,dx$

$\frac{y^3}{3}=x^2-x+c$ ✓

when $x=0$, $y=\sqrt[3]{3}$

So $\frac{(\sqrt[3]{3})^3}{3}=c$ or $c=1$

$\frac{y^3}{3}=x^2-x+1$ ✓

$y^3=3x^2-3x+3$

$y=\sqrt[3]{3x^2-3x+3}$

Q 2 Baulkham

A solution to the differential equation $\frac{dy}{dx}=\frac{2}{\cos(x+y)+\cos(x-y)}$ can be obtained from:

A. $\int \cos y\,dy=\int \sec x\,dx$

B. $\int \sec y\,dy=\int \cos x\,dx$

C. $\int \sin y\,dy=\int \operatorname{cosec} x\,dx$

D. $\int \operatorname{cosec} y\,dy=\int \sin x\,dx$

$$\frac{dy}{dx}=\frac{2}{\cos(x+y)+\cos(x-y)}$$

$$\frac{dy}{dx}=\frac{2}{\cos x\cos y-\sin x\sin y+\cos x\cos y+\sin x\sin y}$$

$$=\frac{2}{2\cos x\cos y}$$

$$=\frac{1}{\cos x\cos y}$$

$$\int \cos y\,dy=\int\frac{1}{\cos x}\,dx$$

$$\int \cos y\,dy=\int \sec x\,dx$$

Q 3 Baulkham

Solve $\frac{dy}{dx}=\frac{y^2}{1+x^2}$ given $y(0)=1$. **3**

$$\frac{dy}{dx}=\frac{y^2}{1+x^2}$$

$$\int_1^y\frac{dy}{y^2}=\int_0^x\frac{dx}{1+x^2}$$

$$\int_1^y y^{-2}\,dy=\int_0^x\frac{dx}{1+x^2}$$

$$\left[\frac{y^{-1}}{-1}\right]_1^y=\left[\tan^{-1}x\right]_0^x$$

$$-\frac{1}{y}-(-1)=\tan^{-1}x$$

$$-\frac{1}{y}+1=\tan^{-1}x$$

$$\frac{1}{y}=1-\tan^{-1}x$$

$$y=\frac{1}{1-\tan^{-1}x}$$

Q 4 Blacktown

Solve the differential equation $\frac{dy}{dx}=\frac{1}{(2-y)\sqrt{2-x^2}}$ given that when $x=1$, $y=0$. Express y as a function of x. **4**

$$\frac{dy}{dx}=\frac{1}{(2-y)\sqrt{2-x^2}}$$

$$(2-y)dy=\frac{1}{\sqrt{2-x^2}}dx$$

$$\int(2-y)dy=\int\frac{1}{\sqrt{2-x^2}}dx$$

$$2y-\frac{y^2}{2}=\sin^{-1}\left(\frac{x}{\sqrt2}\right)+c$$

$x=1, y=0$

$$2\times0-\frac{0^2}{2}=\sin^{-1}\left(\frac{1}{\sqrt2}\right)+c$$

$$0=\frac{\pi}{4}+c$$

$$c=-\frac{\pi}{4}$$

$$2y-\frac{y^2}{2}=\sin^{-1}\left(\frac{x}{\sqrt2}\right)-\frac{\pi}{4}$$

$$y^2-4y=\frac{\pi}{2}-2\sin^{-1}\left(\frac{x}{\sqrt2}\right)$$

$$y^2-4y+4=\frac{\pi}{2}-2\sin^{-1}\left(\frac{x}{\sqrt2}\right)+4$$

$$(y-2)^2=\frac{\pi}{2}+4-2\sin^{-1}\left(\frac{x}{\sqrt2}\right)$$

$$y-2=\pm\sqrt{\frac{\pi}{2}+4-2\sin^{-1}\left(\frac{x}{\sqrt2}\right)}$$

$$y=2\pm\sqrt{\frac{\pi}{2}+4-2\sin^{-1}\left(\frac{x}{\sqrt2}\right)}$$

Since $x=1, y=0$

$$\therefore y=2-\sqrt{\frac{\pi}{2}+4-2\sin^{-1}\left(\frac{x}{\sqrt2}\right)}$$

Q 5 Cheltenham Girls

For the differential equation $y' = \dfrac{6}{5x^2+4x-1}$:

i) Show that $y' = \dfrac{5}{5x-1} - \dfrac{1}{x+1}$. 1

ii) Find the solution to the differential equation; given that when $x = \frac{1}{2}$, $y = 3$. 2

i) $\dfrac{5}{5x-1} - \dfrac{1}{x+1} = \dfrac{5x+5-5x+1}{5x^2+4x-1} = \dfrac{6}{5x^2+4x-1}$

ii)
$$y = \int \frac{6}{5x^2+4x-1}\,dx = \int \frac{5}{5x-1} - \frac{1}{x+1}\,dx$$
$$= \ln(5x-1) - \ln(x+1) + C$$
$$= \ln\left(\frac{5x-1}{x+1}\right) + C$$

when $x = \frac{1}{2}, y = 3$

$$3 = \ln\left(\frac{\frac{5}{2}-1}{\frac{1}{2}+1}\right) + C$$
$$3 = \ln\left(\frac{\frac{3}{2}}{\frac{3}{2}}\right) + C$$
$$3 = \ln(1) + C$$
$$3 = 0 + C$$
$$\therefore C = 3$$
$$y = \ln\left(\frac{5x-1}{x+1}\right) + 3$$

Q 6 Grafton

Which of the following differential equations can **NOT** be solved by separation of the variables?

(A) $\dfrac{dy}{dx} = x$ (B) $\dfrac{dy}{dx} = \dfrac{x}{y}$ (C) $\dfrac{dy}{dx} = xy$ (D) $\dfrac{dy}{dx} = x + y$

$\dfrac{dy}{dx} = x + y$

Q 7 Kambala

How a fish moves through water is described by the differential equation 2

$$\frac{dy}{dx} = 1 + y^2.$$

Solve this equation, given that $(0,0)$ lies on the curve.

$\dfrac{dy}{dx} = 1+y^2$

$\dfrac{dx}{dy} = \dfrac{1}{1+y^2}$

$\int dx = \int \dfrac{1}{1+y^2}\,dy$ ✓ 1 rearrange.

$x = \tan^{-1} y + C$. when $x=0, y=0 \therefore C=0$.

$\therefore x = \tan^{-1} y$

$y = \tan x$. ✓ 1 ans.

Q 8 Manly

Solve the differential equation, 3

$$(x^2+5)\frac{dy}{dx} = xy$$

given that $y(2) = 3$

$$(x^2+5)\frac{dy}{dx} = xy$$
$$\frac{1}{y}\,dy = \frac{x}{x^2+5}\,dx$$
$$\int \frac{1}{y}\,dy = \int \frac{x}{x^2+5}\,dx$$
$$\ln|y| + C_1 = \frac{1}{2}\ln|(x^2+5)|$$
$$2\ln|y| + C_2 = \ln|(x^2+5)|$$

$$x = 2 \quad y = 3$$
$$2\ln 3 + C_2 = \ln(9)$$
$$\ln(3^2) + C_2 = \ln(9)$$
$$C_2 = 0$$
$$\ln|y| = \ln\left(\sqrt{x^2+5}\right)$$
$$|y| = \sqrt{x^2+5}$$
$$y = \pm\sqrt{x^2+5}$$
$$y(2) = 3 \Rightarrow y = \sqrt{x^2+5}$$

Q 9 Manly

The population, P, of animals in an environment in which there are scarce resources is increasing in a manner described by the differential equation $\dfrac{dP}{dt} = P(100 - P)$.

When $t = 0$, $P = 10$.

i) Show that $\dfrac{1}{100}\left(\dfrac{1}{P} + \dfrac{1}{100 - P}\right) = \dfrac{1}{P(100 - P)}$ **1**

ii) Find an expression for P in terms of t. **3**

a-i

$$\frac{1}{100}\left(\frac{1}{P} + \frac{1}{100 - P}\right) = \frac{1}{P(100 - P)}$$

$$\begin{aligned} LHS &= \frac{1}{100}\left(\frac{1}{P} + \frac{1}{100 - P}\right) \\ &= \frac{1}{100}\left[\frac{(100 - P) + P}{P(100 - P)}\right] \\ &= \frac{1}{100} \times \frac{100}{P(100 - P)} \\ &= \frac{1}{P(100 - P)} \end{aligned}$$

a-ii

$$\frac{dP}{dt} = P(100 - P)$$

$$\frac{dt}{dP} = \frac{1}{P(100 - P)}$$

$$dt = \frac{1}{P(100 - P)}\, dP$$

$$\int dt = \int \frac{1}{100}\left(\frac{1}{P} + \frac{1}{100 - P}\right) dP$$

$$\int 100dt = \int \frac{1}{P} + \frac{1}{100 - P}\, dP$$

$$100t + C = [\ln|P| - \ln(|100 - P|)]$$

$$100t + C = \ln \frac{P}{100 - P}$$

$$Ae^{100t} = \frac{P}{100 - P}$$

when $t = 0$, $P = 10$

$$A = \frac{10}{90} = \frac{1}{9}$$

$$\frac{1}{9}e^{100t} = \frac{P}{100 - P}$$

$$e^{100t}(100 - P) = 9P$$

$$100e^{100t} - Pe^{100t} = 9P$$

$$100e^{t} = P(9 + e^{100t})$$

$$P = \frac{100e^{t}}{9 + e^{100t}}$$

Q10 Newington

The population, P, of animals in an environment in which there are scarce resources is increasing such that $\dfrac{dP}{dt} = P(100 - P)$, where t is time.

When $t = 0$, $P = 10$.

(i) Show that $\dfrac{1}{100}\left(\dfrac{1}{P} + \dfrac{1}{100 - P}\right) = \dfrac{1}{P(100 - P)}$. **1**

(ii) Find an expression for P in terms of t. **3**

(i)

$$\frac{1}{100}\left(\frac{1}{P} + \frac{1}{100 - P}\right) = \frac{1}{100}\left(\frac{100 - P + P}{P(100 - P)}\right) = \frac{1}{P(100 - P)}$$

(ii)

$$\frac{dP}{dt} = P(100 - P)$$

$$\frac{1}{P(100 - P)}dP = dt$$

$$\int \frac{1}{P(100 - P)}dP = \int dt$$

$$\int \frac{1}{100}\left(\frac{1}{P} + \frac{1}{100 - P}\right)dP = \int dt$$

$$\int \left(\frac{1}{P} + \frac{1}{100 - P}\right)dP = 100\int dt$$

$$\log_e P - \log_e(100 - P) = 100t + c$$

$$t = 0, P = 10 \Rightarrow \log_e 10 - \log_e 90 = c$$

$$c = \log_e \frac{1}{9}$$

$$\log_e\left(\frac{P}{100 - P}\right) = 100t + \log_e \frac{1}{9}$$

$$\left(\frac{P}{100 - P}\right) = e^{100t + \log_e \frac{1}{9}}$$

$$\frac{P}{100 - P} = \frac{1}{9}e^{100t} \quad \Longrightarrow \quad \frac{100 - P}{P} = 9e^{-100t}$$

$$9P = 100e^{100t} - Pe^{100t}$$

$$9P + Pe^{100t} = 100e^{100t}$$

$$P = \frac{100e^{100t}}{9 + e^{100t}}$$

$$\frac{100}{P} - 1 = 9e^{-100t}$$

$$\frac{100}{P} = 1 + 9e^{-100t}$$

$$P = \frac{100}{1 + 9e^{-100t}}$$

Q11 Normanhurst

Consider the differential equation $\frac{dy}{dx} - \frac{2y}{x} = 0$. Find the equation of the solution curve if it passes through $(1, -1)$. **3**

$$\frac{dy}{dx} = \frac{2y}{x}$$
$$\frac{1}{2}\int \frac{1}{y}dy = \int \frac{1}{x}dx$$
$$\frac{1}{2}\ln|y| = \ln|x| + c$$
$$\because \text{When } x = 1, y = -1,$$
$$c = 0$$
$$\therefore \frac{1}{2}\ln(-y) = \ln x$$
$$\ln(-y) = \ln x^2$$
$$-y = x^2$$
$$y = -x^2$$

Q12 Penrith

(i) Solve the differential equation $\frac{dy}{dx} = \frac{x^2}{y^2}$ *1*

(ii) Find the solution of this equation that satisfies the initial condition $y(0) = 2$. *1*

(i) $\frac{dy}{dx} = \frac{x^2}{y^2}$

$y^2 dy = x^2 dx$ (variable-separable)

integrate both sides,

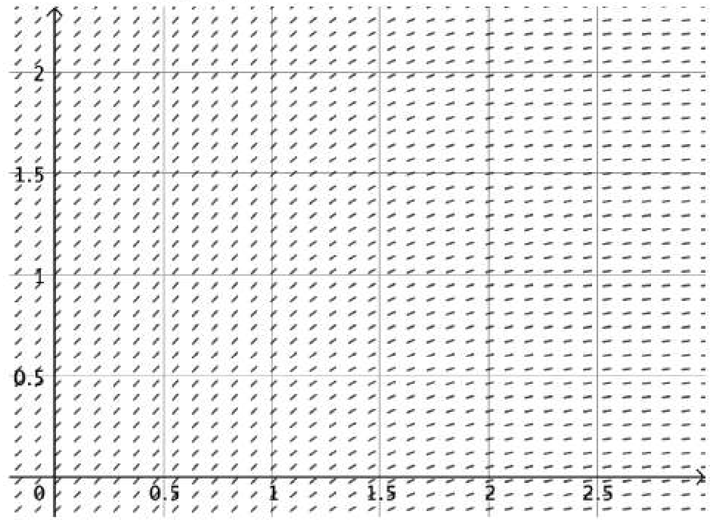
$\frac{y^3}{3} = \frac{x^3}{3} + C$

$\cancel{y^3 = x}$

(ii) $y(0) = 2 \Rightarrow \boxed{C = \frac{8}{3}}$

Q13 Normanhurst

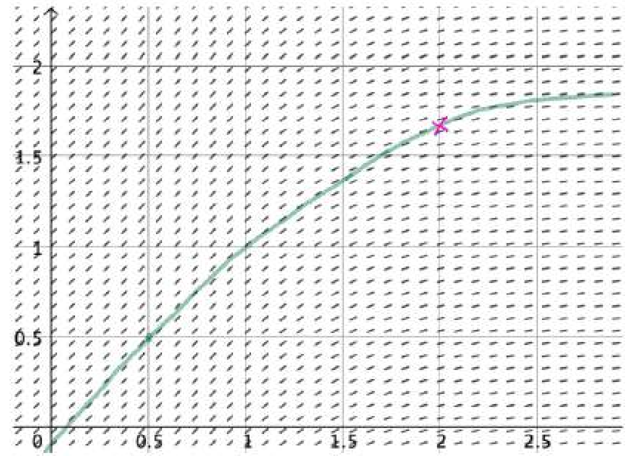
The slope field of $\frac{dy}{dx} = \frac{1+x^2}{1+x^4}$ is shown below.



i. It is given that the solution curve passes through $(0.5, 0.5)$. On the grid provided on page 12, sketch the solution curve on the slope field. **1**

ii. Find the approximate value of y when $x = 2$. Give your answer correct to one decimal place. **1**

i.



ii. for a value in $1.6 \leq y \leq 1.8$

Q14 Prairiewood

(i) Show that $\frac{100}{y(100-y)} = \frac{1}{y} + \frac{1}{100-y}$. 1

(ii) Use integration to show that the equation $\frac{dy}{dx} = \frac{y(100-y)}{100}$, given $y = 20$ when $x = 0$, has solution $y = \frac{100}{1+4e^{-x}}$. 3

(i) $\frac{100}{y(100-y)} = \frac{(100-y)+y}{y(100-y)}$

$= \frac{100-y}{y(100-y)} + \frac{y}{y(100-y)}$

$= \frac{1}{y} + \frac{1}{100-y}$

(ii) $\frac{dy}{dx} = \frac{y(100-y)}{100}$

$\frac{dx}{dy} = \frac{1}{y} + \frac{1}{100-y}$

$x = \ln|y| - \ln|100-y| + C$

$x = \ln\left|\frac{y}{100-y}\right| + C$

$x=0,\ y=20 \Rightarrow C = -\ln\frac{20}{80} = \ln 4$

$x = \ln\left(\frac{4y}{100-y}\right)$

$e^{-x} = \frac{100-y}{4y}$

$e^{-x} = \frac{1}{4}\left(\frac{100}{y} - 1\right)$

$4e^{-x} = \frac{100}{y} - 1$

$\frac{100}{y} = 1 + 4e^{-x}$

But $y = 20$ when $x = 0$

$\frac{100}{y} = 1 + 4e^{-x}$

$\Rightarrow y = \frac{100}{1+4e^{-x}}$

Q15 Prairiewood

A curve has gradient given by $\frac{dy}{dx} = \frac{x}{y}$. Find an expression for y in terms of x given that the curve passes through the point $(0,1)$. 2

$\frac{dy}{dx} = \frac{x}{y}$

$\int y \cdot dy = \int x \cdot dx$.

$\frac{y^2}{2} = \frac{x^2}{2} + C$.

$y^2 = x^2 + C$. pass $(0,1)$

$\Rightarrow C = 1$

$y^2 = x^2 + 1$

Q16 Sydney Boys

Given the relation $4x^2 + 9y^2 = 36$:

(i) Show that $\frac{dy}{dx} = -\frac{4x}{9y}$. 1

(ii) Find the gradient of the tangent at the point on the curve where $x = 2$, and $y > 0$. 1

i) $4x^2+9y^2=36$

$\frac{d}{dx}(4x^2) + \frac{d}{dx}(9y^2) = \frac{d}{dx}(36)$

$\frac{d}{dx}(4x^2) + \frac{dy}{dx} \times \frac{d}{dy}(9y^2) = \frac{d}{dx}(36)$

$8x + \frac{dy}{dx} \times 18y = 0$

$18y\frac{dy}{dx} = -8x$

$\frac{dy}{dx} = \frac{-8x}{18y} = \frac{-4x}{9y}$ as required

ii) Let $x = 2$

$4 \times 2^2 + 9y^2 = 36$

$9y^2 = 20$

$y^2 = \frac{20}{9}$

$y = \sqrt{\frac{20}{9}} = \frac{2\sqrt{5}}{3}$ (since $y>0$)

$\frac{dy}{dx} = \frac{-4 \times 2}{9 \times \frac{2\sqrt{5}}{3}}$ (or $\frac{-8}{6\sqrt{5}}$, $-\frac{4}{3\sqrt{5}}$, $-\frac{4\sqrt{5}}{15}$, -0.596 or equivalent)

Q17 Sydney Grammar

Solve the initial value problem where $\frac{dy}{dx} = 2xe^{-y}$ given $y(0) = \ln 4$. Express your answer with y as the subject. [3]

b) $\frac{dy}{dt} = 2xe^{-y}$ where $y(0) = \ln 4$

No constant soln $e^{-y} \neq 0$

Separate Variables $\int e^{y} dy = \int 2x dx$

$e^{y} = x^2 + C$

$y = \ln|x^2 + c|$

$y(0) = \ln 4$ $\therefore$ $\ln 4 = \ln|c|$

$c = 4$

$\therefore$ $y = \ln|x^2 + 4|$

Q18 Sydney Tech

The number of elephants, N, in a population at time t is given by $N = Ae^{kt} + 750$, with constants $A > 0$ and $k > 0$. Which of the following is the correct differential equation?

(A) $\frac{dN}{dt} = -k(N + 750)$

(B) $\frac{dN}{dt} = k(N + 750)$

(C) $\frac{dN}{dt} = -k(N - 750)$

(D) $\frac{dN}{dt} = k(N - 750)$

Q19 Sydney Tech

Find the particular solution to $\frac{dy}{dx} = \frac{2x}{3y^2}$ where $y(0) = 1$ 3

$\frac{dy}{dx} = \frac{2x}{3y^2}$

$\int 3y^2 \, dy = \int 2x \, dx$

$y^3 = x^2 + c$

At $x = 0$, $y = 1$

$1 = 0 + c$

$c = 1$

So $y^3 = x^2 + 1$

$y = \sqrt[3]{x^2 + 1}$

Q20 Total Education

The differential equation, $y' = 3\sin x + 2\cos x$, has a solution on the curve $y = f(x)$ at $P(\pi, 5)$.

What is the value of the arbitrary constant?

A. 1 B. 2 C. 3 D. 4

$y' = 3\sin x + 2\cos x$

$\int y' \, dx = \int 3\sin x + 2\cos x \, dx$

$y = -3\cos x + 2\sin x + C$

at $(\pi, 5) \Rightarrow 5 = (-3)(-1) + 2(0) + C$

$\Rightarrow C = 2$

Q21 Total Education

(i) Solve the differential equation $\frac{dy}{dx} = \frac{y(x-1)}{x}$ using separation of variables. **3**

(ii) If $y(1) = 2$, find the value of the constant. **1**

(iii) Hence, find the value of y if $x = 2$. **1**

(i) $\frac{dy}{dx} = \frac{y(x-1)}{x}$

$\Rightarrow \int \frac{dy}{y} = \int \frac{x-1}{x}\,dx \Rightarrow \ln y = \int 1 - \frac{1}{x}\,dx$

$= x - \ln x + C$

$\therefore y = e^{x - \ln x + c}$

$= \frac{e^x e^c}{e^{\ln x}}$

$= \frac{Ae^x}{x}$

(ii) $y(1) = 2 \Rightarrow 2 = \frac{Ae^1}{1}$

$\therefore A = \frac{2}{e}$

$\therefore y = \frac{2e^x}{ex}$

(iii) $x = 2 \Rightarrow y = \frac{2e^2}{2e} = e$

Q22 CSSA

Solve the equation $\sec x \frac{dy}{dx} = \frac{e^{\sin x}}{y}$, given $x = 0$ when $y = 0$. **3**

$\sec x \frac{dy}{dx} = \frac{e^{\sin x}}{y}$

$y\,dy = \cos x e^{\sin x}\,dx$

integrating both sides:

$\int y\,dy = \int \cos x e^{\sin x}\,dx$

$\frac{y^2}{2} = e^{\sin x} + c$ since $\frac{d}{dx}\left(e^{\sin x}\right) = \cos x e^{\sin x}$

given $x = 0$, $y = 0$

$\frac{0^2}{2} = e^{\sin 0} + c$

$c = -1$

$\therefore y^2 = 2e^{\sin x} - 2$

Q23 Enterprise

The solution to the differential equation

$$\frac{dy}{dx} = 9x^2 + 2x - 1$$

Given that $y = 2$ when $x = 1$ is

(A) $y = 3x^3 + x^2 - x + 1$

(B) $y = 3x^3 + x^2 - x - 2$

(C) $y = 3x^3 + x^2 - x + 10$

(D) $y = 3x^3 + x^2 - x - 1$

Q1 — ASCHAM

A population of 1200 feral cats is released into Darling Point. The rate of increase of the feral cat population is: $\frac{dP}{dt}=\frac{P}{30}\left(1-\frac{P}{10000}\right)$ where P is the feral cat population and t is the number of months.

(i) Show that $\frac{10000}{P(10000-P)}=\frac{1}{P}+\frac{1}{10000-P}$ **1**

(ii) Hence solve the differential equation $\frac{dP}{dt}=\frac{P}{30}\left(1-\frac{P}{10000}\right)$ for P in terms of t. **3**

(iii) Hence find the limiting feral cat population. **1**

(iv) How many months does it take for the population to double? **1**

(i) $\frac{10000}{P(10000-P)}=\frac{1}{P}+\frac{1}{10000-P}$

RHS $=\frac{1}{P}+\frac{1}{10000-P}=\frac{(10000-P)+P}{P(10000-P)}$ ✓

$=\frac{10000}{P(10000-P)}$

$=$ LHS

(ii) $\frac{dP}{dt}=\frac{P}{30}\left(1-\frac{P}{10000}\right)=\frac{P}{30}\left(\frac{10000-P}{10000}\right)$

$\int\frac{10000}{P(10000-P)}\,dP=\int\frac{1}{30}\,dt$ ✓

$\int\left(\frac{1}{P}+\frac{1}{10000-P}\right)dP=\frac{1}{30}t+c$

$\ln P-\ln(10000-P)=\frac{1}{30}t+c$ ✓

when $t=0$, $P=1200$

$\ln 1200-\ln(10000-1200)=c$

$\therefore c=\ln\left(\frac{1200}{8800}\right)$

$=\ln\frac{3}{22}$

$\ln P-\ln(10000-P)=\frac{1}{30}t+\ln\frac{3}{22}$

$\ln\left(\frac{P}{10000-P}\right)=\frac{1}{30}t+\ln\frac{3}{22}$ ✓

$\frac{P}{10000-P}=e^{\left(\frac{1}{30}t+\ln\left(\frac{3}{22}\right)\right)}$

$P=\left(e^{\left(\frac{1}{30}t\right)}\cdot e^{\ln\left(\frac{3}{22}\right)}\right)(10000-P)$

$=\frac{3}{22}e^{\frac{1}{30}t}(10000)-\frac{3}{22}e^{\frac{1}{30}t}(P)$

$P+\frac{3}{22}e^{\frac{1}{30}t}P=\frac{3}{22}e^{\frac{1}{30}t}(10000)$

$P\left(1+\frac{3}{22}e^{\frac{1}{30}t}\right)=\frac{30000}{22}e^{\frac{1}{30}t}$

$P=\frac{30000}{22}\,\frac{e^{\frac{1}{30}t}}{\left(1+\frac{3}{22}e^{\frac{1}{30}t}\right)}\begin{matrix}\div e^{\frac{1}{30}t}\\ \div e^{\frac{1}{30}t}\end{matrix}$

$P=\frac{30000}{22\left(e^{-\frac{1}{30}t}+\frac{3}{22}\right)}$ ✓

(iii) as $t\to\infty$, $P\to\frac{30000}{22\times\frac{3}{22}}=10000$ ✓

(iv) when $t=0$, $P=1200$

population doubles when $P=2400$

$2400=\frac{30000}{22\left(e^{-\frac{1}{30}t}+\frac{3}{22}\right)}$

$e^{-\frac{1}{30}t}+\frac{3}{22}=\frac{30000}{22\times 2400}=\frac{25}{22\times 2}$

$e^{-\frac{1}{30}t}=\frac{25}{22\times 2}-\frac{6}{22\times 2}$

$=\frac{19}{44}$

$e^{\frac{1}{30}t}=\frac{44}{19}$

$\frac{1}{30}t=\ln\left(\frac{44}{19}\right)$

$t=30\ln\left(\frac{44}{19}\right)$

$=25.19\ldots$

$=25$ months (to the nearest month) ✓

Q 2 BAULKHAM

A team of biologists released 500 fish into a lake with a maximum carrying capacity of 10000 fish. It was found that the number of fish tripled during the first year. The fish population, N, in the lake after t years is modelled by the logistic equation:

$$\frac{dN}{dt} = kN(10000 - N) \qquad \text{where } k \text{ is a constant.}$$

(i) Given $\dfrac{10000}{N(10000-N)} = \dfrac{1}{N} + \dfrac{1}{10000-N}$, solve the differential equation to show that the fish population after t years is $N = \dfrac{10000}{1+19e^{-10000kt}}$. **3**

(ii) Show that $k = \dfrac{1}{10000}\ln\left(\dfrac{57}{17}\right)$. **2**

(iii) Correct to the nearest month, how long will it take for the fish population in the lake to reach 7000 fish? **2**

(i)

$$\frac{dN}{dt} = kN(10000-N)$$

$$\frac{10000\,dN}{N(10000-N)} = 10000k\,dt$$

$$\int_{500}^{N} \frac{10000\,dN}{N(10000-N)} = 10000k\int_0^t dt$$

$$\int_{500}^{N} \frac{1}{N} + \frac{1}{10000-N}\,dN = 10000k\int_0^t dt$$

$$\left[\ln N - \ln(10000-N)\right]_{500}^{N} = 10000k\left[t\right]_0^t$$

$$\left[\ln\left(\frac{N}{10000-N}\right)\right]_{500}^{N} = 10000kt$$

$$\ln\left(\frac{N}{10000-N}\right) - \ln\left(\frac{500}{10000-500}\right) = 10000kt$$

$$\ln\left(\frac{19N}{10000-N}\right) = 10000kt$$

$$\ln\left(\frac{10000-N}{19N}\right) = -10000kt$$

$$\frac{10000-N}{19N} = e^{-10000kt}$$

$$10000 - N = 19Ne^{-10000kt}$$

$$10000 = N + 19Ne^{-10000kt}$$

$$N = \frac{10000}{1+19e^{-10000kt}}$$

(ii)

$$N = \frac{10000}{1+19e^{-10000kt}}$$

$$1500 = \frac{10000}{1+19e^{-10000k}}$$

$$1+19e^{-10000k} = \frac{20}{3}$$

$$19e^{-10000k} = \frac{17}{3}$$

$$e^{-10000k} = \frac{17}{57}$$

$$-10000k = \ln\left(\frac{17}{57}\right)$$

$$k = \frac{1}{10000}\ln\left(\frac{57}{17}\right)$$

(iii)

$$N = \frac{10000}{1+19e^{-10000kt}}$$

$$7000 = \frac{10000}{1+19e^{\ln\left(\frac{17}{57}\right)t}}$$

$$7000 = \frac{10000}{1+19\times\left(\frac{17}{57}\right)^t}$$

$$1+19\times\left(\frac{17}{57}\right)^t = \frac{10}{7}$$

$$19\times\left(\frac{17}{57}\right)^t = \frac{3}{7}$$

$$\left(\frac{17}{57}\right)^t = \frac{3}{133}$$

$$t\ln\left(\frac{17}{57}\right) = \ln\left(\frac{3}{133}\right)$$

$$t = \ln\left(\frac{3}{133}\right) \div \ln\left(\frac{17}{57}\right)$$

$= 3.134$ years (3 d.p.)

$= 38$ months (nearest month)

or 3 years and 2 months (nearest month)

Q 3 KILLARA

Two chemicals A and B are poured into a mixing machine. Initially the machine has 40 L of chemical A. Then chemical A is poured in at a rate of 2 L/min and at the same time chemical B is poured in at 6 L/min. The mixing machine constantly mixes the chemicals and the mixture flows out at 4 L/min.

(i) Show that the expression for the rate of change of the volume of chemical B in the mixing machine t minutes after the pouring commences is given by: **1**

$$\frac{dB}{dt} = 6 - \frac{B}{10+t}$$

(ii) Find values of m and n, real numbers, such that **2**

$$B(t) = \frac{mt^2 + nt}{10+t}$$

is a solution to the differential equation in part (i).

(iii) A useable mixture is roughly two parts chemical A to three parts chemical B. How many minutes should elapse before the outflow is the required mix? **2**

(i) Total Volume of liquid in the tank at time t = $(40 + 2t + 6t) - (4t) = 40 + 4t$ Litres

Proportion of B in the container =

$$\frac{B}{40+4t}$$

Amount of B flowing out per minute =

$$\frac{B}{40+4t} \times 4 =$$

$$\frac{B}{10+t} \; L/min$$

$$\frac{dB}{dt} = \text{Rate of inflow of B} - \text{rate of outflow of B}$$

$$= 6\,L/\min - \frac{B}{10+t}\,L/min$$

$$\frac{dB}{dt} = 6 - \frac{B}{10+t}$$

(ii)

$$B = \frac{mt^2+nt}{10+t}$$

$$\frac{dB}{dt} = \frac{(10+t)(2mt+n) - (mt^2+nt)}{(10+t)^2}$$

$$= \frac{2mt+n}{10+t} - \frac{mt^2+nt}{(10+t)^2}$$

Equating,

$$\frac{2mt+n}{10+t} - \frac{mt^2+nt}{(10+t)^2} = 6 - \frac{B}{10+t}$$

$$\frac{2mt+n}{10+t} = 6$$

$$2mt + n = 60 + 6t$$

$$n = 60, \quad 2m = 6, \text{ ie. } m = 3$$

(iii)

$$B = \frac{3t^2+60t}{10+t}$$

$$\frac{B}{40+4t} = \frac{3}{5}$$

$$\frac{3t^2+60t}{4(10+t)^2} = \frac{3}{5}$$

$$\frac{t^2+20t}{4(10+t)^2} = \frac{1}{5}$$

$$5t^2 + 100t = 400 + 80t + 4t^2$$

$$t^2 + 20t - 400 = 0$$

$$t = 10\sqrt{5} - 10 \approx 12.36 \quad t > 0$$

Q 4 NEAP

The area A cm^2 is occupied by a bacterial colony. The colony has its growth modelled by the logistic equation $\frac{dA}{dt} = \frac{1}{25}A(50 - A)$ where $t \geq 0$ and t is measured in days. At time $t = 0$, the area occupied by the bacteria colony is $\frac{1}{2}$ cm^2.

(i) Show that $\frac{1}{A(50-A)} = \frac{1}{50}\left(\frac{1}{A} + \frac{1}{50-A}\right)$. **1**

(ii) Using the result from part (b) (i), solve the logistic equation and hence show that **3**

$$A = \frac{50}{1 + 99e^{-2t}}.$$

(iii) According to this model, what is the limiting area of the bacteria colony? **1**

(iv) Find the exact time when the rate of change in the area occupied by the bacterial colony is at its maximum. **2**

(i) Start with the RHS and show that it equals the LHS.

$$\text{RHS} = \frac{1}{50}\left(\frac{(50-A)+A}{A(50-A)}\right)$$

$$= \frac{1}{A(50-A)}$$

$$= \text{LHS}$$

(ii) This is a differential equation of the form $\frac{dA}{dt} = g(A)$.

Attempt to separate variables and integrate both sides.

$$\int 1\,dt = \int \frac{25}{A(50-A)}\,dA$$

$$t = \frac{1}{2}\int\left(\frac{1}{A} + \frac{1}{50-A}\right)dA \text{ (using the part (i) result)}$$

$$= \frac{1}{2}(\ln|A| - \ln|50-A|) + c$$

$$= \frac{1}{2}\ln\left|\frac{A}{50-A}\right| + c$$

Rearranging gives $A_0 e^{2t} = \frac{A}{50-A}$ where $A_0 = e^{-2c}$ and hence $A_0 > 0$.

When $t = 0$, $A = \frac{1}{2}$ and so $A_0 = \frac{1}{99}$.

Note: There are various possible ways to find the value of the constant.

$$e^{2t} = \frac{99A}{50-A}$$

$$99Ae^{-2t} = 50 - A$$

$$A(1 + 99e^{-2t}) = 50$$

$$\text{So } A = \frac{50}{1+99e^{-2t}}.$$

(iii) As $t \to \infty$, $1 + 99e^{-2t} \to 1$ and so $A \to \frac{50}{1} = 50$.

The limiting area of the bacteria colony is 50 cm^2.

Q 5 SYDNEY BOYS

A model for the population, K, of kangaroos is

$$K = \frac{21\,000}{7+3e^{-\frac{t}{3}}},$$

where t is the time in years from today.

(i) Show that K satisfies the differential equation **2**

$$\frac{dK}{dt} = \frac{1}{3}\left(1 - \frac{K}{3000}\right)K.$$

(ii) What is the population today? **1**

(iii) What does the model predict that the eventual population will be? **1**

(iv) What is the annual percentage rate of growth today? **1**

(a) (i) $K = \frac{21000}{7+3e^{-\frac{t}{3}}} = 21000(7+3e^{-\frac{t}{3}})^{-1}$ Let $u = 7 + e^{-\frac{t}{3}}$, so $\frac{du}{dt} = -e^{-\frac{t}{3}}$

$K = 21000u^{-1}$, $\quad \frac{dK}{du} = -\frac{21000}{u^2}$

$\frac{dK}{dt} = \frac{dK}{du} \times \frac{du}{dt}$ (chain rule)

$= \frac{21000}{e^{-\frac{t}{3}}(7+3e^{-\frac{t}{3}})^2}$

1 mark for differentiating

$\text{RHS} = \frac{1}{3}\left(1 - \frac{K}{3000}\right)K$

$= \frac{1}{3}\left(1 - \frac{7}{7+3e^{-\frac{t}{3}}}\right)\left(\frac{21000}{7+3e^{-\frac{t}{3}}}\right)$

$= \left(\frac{7+3e^{-\frac{t}{3}}-7}{7+3e^{-\frac{t}{3}}}\right)\left(\frac{7000}{7+3e^{-\frac{t}{3}}}\right)$

$= \frac{21000e^{-\frac{t}{3}}}{(7+3e^{-\frac{t}{3}})^2}$

$= \frac{dK}{dt}$

= LHS.

1 mark for matching differential equation with derivative

(ii) Sub. in $t = 0$: $K = \frac{21000}{7+3e^0} = 2100$

1 mark

(iii) Find the limit as $t \to \infty$: $K = \frac{21000}{7+3e^{-\frac{t}{3}}}$ $\quad (\lim_{t\to\infty} e^{-t} = 0)$

$= \frac{21000}{7}$

$= 3000$

1 mark

(iv) At $t = 0$: $\frac{dK}{dt} = \frac{21000}{e^0(7+3e^0)^2} = \frac{21000}{100} = 210$ kangaroos/year

½ mark

So $\frac{210}{2100} \times 100\% = 10\%$ annual rate.

½ mark

Q 6 SYDNEY GRAMMAR

Data suggests that the number of cases of infection from a particular disease tends to fluctuate between two values over a period of approximately six months.

Let P be the number of cases after t months, where P is measured in thousands. Initially there are 1000 cases.

(i) Suppose that P is modelled by the equation: **[2]**

$$P = \frac{2}{2 - \sin t}.$$

Verify that P satisfies the differential equation $\frac{dP}{dt} = \frac{1}{2}P^2 \cos t$.

(ii) An alternative model is proposed with a different differential equation: **[4]**

$$\frac{dP}{dt} = \frac{1}{2}(2P^2 - P)\cos t.$$

Use the result that $\frac{1}{P(2P-1)} = \frac{2}{2P-1} - \frac{1}{P}$ to solve this differential equation, showing that:

$$P = \frac{1}{2 - e^{\frac{1}{2}\sin t}}$$

and that this new solution also satisfies the initial condition.

(iii) Find the greatest and least values of P predicted by both models for $t \geq 0$. Give your answers in exact form and then rounded to three decimal places to enable easy comparison between the two models. **[2]**

b)(i) $P = \frac{2}{2-\sin t}$

$= 2(2-\sin t)^{-1}$ ✓

$\frac{dP}{dt} = -2(2-\sin t)^{-2} \times (-\cos t)$ ✓

$= \frac{2\cos t}{(2-\sin t)^2}$

$= \frac{4}{(2-\sin t)^2} \times \frac{\cos t}{2}$ ✓ "SHOW"

$= \frac{P^2 \cos t}{2}$ (as required)

(ii) $\frac{dP}{dt} = \frac{1}{2}P(2P-1)\cos t$

Constant Solns $P=0$ and $P=\frac{1}{2}$

$\int \frac{1}{P(2P-1)}\,dP = \int \frac{\cos t}{2}\,dt$

$\int \frac{-1}{P} + \frac{2}{2P-1}\,dP = \frac{\sin t}{2} + c$ ✓

$\ln|2P-1| - \ln|P| = \frac{1}{2}\sin t + c$ "SHOW"

$\ln\left|\frac{2P-1}{P}\right| = \frac{1}{2}\sin t + c$

$\left|\frac{2P-1}{P}\right| = e^{\frac{1}{2}\sin t} \times e^{c}$

$= Ae^{\frac{1}{2}\sin t}$ ✓ $(A>0)$

$\frac{2P-1}{P} = Ae^{\frac{1}{2}\sin t}$ $(A>0$ or $A<0)$

then include constant soln $P=\frac{1}{2}$ ie $A=0$

$2P-1 = APe^{\frac{1}{2}\sin t}$

$P(2-Ae^{\frac{1}{2}\sin t}) = 1$ ✓

$P = \frac{1}{2-Ae^{\frac{1}{2}\sin t}}$ ✓

but $t=0$ $P=1$ $\therefore 1 = \frac{1}{2-Ae^0} \Rightarrow A=1$ ✓

$\therefore P = \frac{1}{2-e^{\frac{1}{2}\sin t}}$ ✓

(iii) $P = \frac{2}{2-\sin t}$

$-1 \leq \sin t \leq 1$

$3 \geq 2-\sin t \geq 1$

$\therefore \frac{2}{3} \leq P \leq \frac{2}{1}$ ✓ ie $0.667 \leq P \leq 2$

Alternate Model

$P = \frac{1}{2-e^{\frac{1}{2}\sin t}}$

$-\frac{1}{2} \leq \frac{1}{2}\sin t \leq \frac{1}{2}$

$e^{-\frac{1}{2}} \leq e^{\frac{1}{2}\sin t} \leq e^{\frac{1}{2}}$

$2-e^{\frac{1}{2}} \leq 2-e^{\frac{1}{2}\sin t} \leq 2-e^{-\frac{1}{2}}$

$\frac{1}{2-\sqrt{e}} \geq P \geq \frac{1}{2-\frac{1}{\sqrt{e}}}$

$\frac{1}{2-\frac{1}{\sqrt{e}}} \leq P \leq \frac{1}{2-\sqrt{e}}$ ✓

$0.718 \leq P \leq 2.847$

Q 7 SYDNEY TECH

25 quokkas are introduced to an island to promote the survival of the species. The growth of their population, Q, over t years, can be modelled by $\frac{dQ}{dt} = 0.0006Q(15\,000 - Q)$.

i. Show that $\frac{1}{0.0006Q(15\,000-Q)} = \frac{1}{9}\left(\frac{1}{Q} + \frac{1}{15\,000-Q}\right)$ **1**

ii. Hence, solve $\frac{dQ}{dt} = 0.0006Q(15\,000 - Q)$, using integration, to show that $Q = \frac{15\,000}{1+Be^{-9t}}$, where B is some constant. **3**

iii. Find the value of B, and hence use the model to estimate the number of quokkas on the island after two months. **2**

iv. What is the maximum number of quokkas that the island will support? **1**

v. After how many months is the number of quokkas to greater than half of the maximum amount? **2**

a) $RHS = \frac{1}{9}\left(\frac{1}{Q} + \frac{1}{15000-Q}\right)$

$= \frac{1}{9}\left(\frac{15000-Q+Q}{Q(15000-Q)}\right)$

$= \frac{15000}{9}\left(\frac{1}{Q(15000-Q)}\right)$

$= \frac{1}{\frac{9}{15000}Q(15000-Q)}$

$= \frac{1}{0.0006Q(15000-Q)}$

$= LHS$

ii/ Hence $\frac{dt}{dQ} = \frac{1}{0.0006Q(15000-Q)}$

$\frac{dt}{dQ} = \frac{1}{9}\left(\frac{1}{Q} + \frac{1}{15000-Q}\right)$

$t = \frac{1}{9}\left(\int \frac{1}{Q}\,dQ - \int \frac{-1}{15000-Q}\,dQ\right)$

$t = \frac{1}{9}\left(\ln|Q| - \ln|15000-Q|\right) + c$

$9(t-c) = \ln\left|\frac{Q}{15000-Q}\right|$

$\therefore \left|\frac{Q}{15000-Q}\right| = e^{9t-9c}$

$\frac{Q}{15000-Q} = Ae^{9t}$, where $A = \pm e^{-9c}$

$Q = 15000Ae^{9t} - QAe^{9t}$

$Q(1+Ae^{9t}) = 15000Ae^{9t}$

$Q = \frac{15000Ae^{9t}}{1+Ae^{9t}}$

$Q = \frac{15000}{\frac{1}{Ae^{9t}}+1}$

$Q = \frac{15000}{1+Be^{-9t}}$, where $B = \frac{1}{A}$

iii/ At $t=0$, $Q=25$

$25 = \frac{15000}{1+Be^{-9\times0}}$

$25 + 25B = 15000$

$25B = 14975$

$B = 599$

$\therefore$ At $t = \frac{2}{12}$ (t is in years)

$Q = \frac{15000}{1+599e^{-9\times\frac{2}{12}}}$

$= 111.395$

$\therefore$ Estimate of 111 quokkas after 2 months.

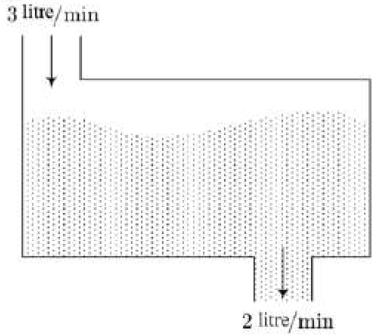
iv/ Carrying capacity as $t \to \infty$

$\therefore Q \to \frac{15000}{1+0} = 15000$

$\therefore$ The maximum number supported is 15000.

Q 8 JAMES RUSE

A tank contains a saltwater solution consisting initially of 20 kg of salt dissolved into 10 L of water. Fresh water is being poured into the tank at a rate of 3 L/min and the solution (kept uniform by stirring) is flowing out at 2 L/min.



(i) Show that Q, the amount of salt (in kilograms), at time t (in minutes) satisfies the equation **2**

$$\frac{dQ}{dt} = -\frac{2Q}{(10+t)}$$

(ii) Solve the differential equation given in (i) to find the amount of salt in the tank after 5 minutes. Answer in kilograms correct to 2 decimal places. **3**

(i)

Implicit assumption is that addition of salt contributes to change in volume negligibly.

Looking for $\frac{dQ}{dt}$, the change in **mass** of salt per unit time.

Now, change in salt in system over time δt is

$$\delta Q = (\text{concentration at time } t) \times (\text{volume out})$$

$$= \frac{Q}{V} \times \delta V$$

$$= \frac{Q}{V}\left(-\frac{2L}{\text{min}}\right)\delta t \ldots (1)$$

Now,

$$\delta V = (V_{\text{in}} - V_{\text{out}}) = \left(3\frac{\text{L}}{\text{min}} - 2\frac{\text{L}}{\text{min}}\right)\delta t = 1\text{L/min } \delta t$$

So

$$\frac{dV}{dt} = \frac{1\text{L}}{\text{min}} \rightarrow V = \left(\frac{1\text{L}}{\text{min}}\right)t + C$$

When $t = 0$ minutes, $V = 10$ L, so $C = 10$ L.

Hence

$$V = 1\frac{\text{L}}{\text{min}}t + 10\,\text{L} \ldots (2)$$

Combining (1) and (2):

$$\delta Q = \frac{Q}{\left(1\frac{\text{L}}{\text{min}}t + 10\,\text{L}\right)}\left(-\frac{2L}{\text{min}}\right)\delta t$$

$$\frac{\delta Q}{\delta t} = -\frac{2Q}{t\frac{\text{L}}{\text{min}} + 10\,\text{L}}$$

Assuming the units now implied,

$$\frac{dQ}{dt} = \lim_{\delta t \to 0}\frac{\delta Q}{\delta t} = -\frac{2Q}{10+t}$$

(ii)

The equation is separable, and assuming the units of Q to be kg, we have

$$\int_{20}^{Q}\frac{dQ}{Q} = -2\int_0^5 \frac{dt}{10+t}$$

$$\log Q - \log 20 = -2(\log(15) - \log 10)$$

$$\log Q = \log\frac{80}{9}$$

So, $Q \approx 8.89$ kg.